MARIAN BOYS' AND GIRLS' SECONDARY SCHOOL
FORM ONE TERMINAL EXAMINATIONS JUNE 2012
BIOLOGY

TIME: $2^1/2$ HRS. **NAME ……**

INSTRUCTIONS

- **This paper consists of section A, B and C**
- **Answer ALL questions**
- **All answer must be written in the space provided in each question**
- **Neatness is important**

SECTION A: (40 marks)

1. The following are multiple choice questions. Choose the correct answer and write its letter in the table provided below

   (i) Which of the following is not a characteristic of an hypothesis
   a) A guess about cause and effects
   b) A deductive inference
   c) A generalization from specific observation
   d) An inductive inference.

   (ii) An object to be observed under the microscope is placed on the
   a) Base
   b) Course adjustment knob
   c) Eye piece
   d) Stage

   (iii) Which is the most accurate definition of a theory?
   a) A tentative cause statement about an observation
   b) An explanatory framework that inter
   c) An initial observation of a natural process
   d) A hypothetical explanation for a natural event.

   (iv) Slips and falls can be prevented by
   a) Making the floor free from peels
   b) Keeping the floor dry
   c) Use the sturdy ladder when climbing to the roof
   d) All the above

   (v) In providing the first aid to a person biten by a snake, all of the following are important except
   a) Assisting the patient to walk to the nearby hospital
   b) Identifying the type of the snake
   c) Tying with bandage above the bitten part
   d) Making a small cut on a biten part

   (vi) Which one of the following is the most correct meaning of a graze?
   a) A minor cut resulted from such sharp objects as razor blades

b) A scrapping off the skin surface which results into profound bleeding
c) A scrapping off the skin surface which results into slight bleeding and injuries
d) A tender swelling resulted from bleeding under the skin
e) Bleeding caused by high blood pressure.

(vii) Which one among the following is not a quality of good first aider.
a) Being able to offer help to another first aider already in control
b) Giving priority to the most urgent conditions
c) Being intelligent by preventing further injury to ensure safety of the injured person
d) Able to act quickly, calm and methodically without reassuring the victim

(viii) If the magnifying power of the eye piece of the compound microscope is X15 and that of high power objective lens is X40. What magnification of a specimen under observation?
a) X600 b) X2.667 c) X25 d) X599

(ix) Movement of curvature is the type of movement experienced by:
a) Animal b) Fishes c) Plants d) Both fishes and plants

(x) Swelling under the skin a bubble filled with liquid is reffered to as
a) Blister b) bruise c) Wound d) buster

(xi) The process involving breakdown of complex foodstuffs into simpler food materials, is termed as: (a) absorption (b) assimilation (c) digestion (d) respiration

(xii) All living matter produces wastes. The process by which wastes are removed from the body is called
(a) excretion (b) sweating (c) Respiration (d) Gutation

(xiii) A scientific term for a complete, individual thing is
a) Modified animal (b) System (c) Organism (d) Modified plant

(xiv) The ability of living things to respond to an external stimulus is reffered to as:

a) Sensitivity
b) Irritability
c) Both A and B are correct
d) Neither A nor B is correct

(xv) Biodegradable wastes are wastes that can be destroyed by:

a) Microbes
b) Heat
c) Chemical changes
d) Human activities

Item no	i	ii	iii	iv	v	vi	vii	viii	ix	x	xi	xii	xiii	xiv	xv
Answer															

2. Match the items in column A with those in column B by writing the letter against the item number in column A in the table drawn below.

Column A		Column B
i)	An irreversible increase in size of an organism	A. Problem identification
ii)	Brings hope and remove fear for death to accident victims	B. Hypothesis formaulation
iii)	Caused by insufficient supply of fresh air	C. Choking
iv)	Producing offspring	D. Hiccups
v)	Injuries to the joint of the body	E. Light microscope
vi)	Food or object get stuck in the throat	F. Iectran microscope
vii)	Specimen viewed as shadow on a fluorescent screen	G. Measuring volumes
viii)	Graduated beaker	H. Measures weight
ix)	The study of the body forms of n organism	I. First aid
x)	"an outbreak if cholera in ujamaa village	J. First aid kit
		K. Suffocation
		L. Fainting
		M. Development
		N. Growth
		O. Morphology
		P. Anatomy
		Q. Sexual reproduction
		R. Reproduction
		S. Fractures
		T. Sprains

Item No	i	ii	iii	**iv**	**v**	**vi**	**vii**	**viii**	**ix**	**x**
Answer										

SECTION B (50 MARKS)

3. For each of the following statements, write **TRUE** or **FALSE** if the statement is true or false respectively

(i) Genetics is the study of inheritance and variations_____

(ii) Something stuck in the throat, drowning and fainting all are the causes for breathing to stop_____

(iii) Bad news, shock, illness, excessive bleeding and unpleasant sight are the causes of fainting_____

(iv) Fire accidents may be prevented by checking the gas electricity systems very occasionally _____

(v) Laying the person down with the affected ear facing upwards and then pouring warm water or oil is one of the recommended first aid to a person whose ear has entered an insect _____

(vi) A person who has swallowed toxic chemical eg. Alkalis, strong acids, etc should be made to vomit_____

(vii) Burns are caused by moist heat in the form of a hot liquid whereas scalds are caused by contact with dry heat_____

(viii) Incision and sucking the snake bitten part may prove more harmful due to introduction of infection, increased bleeding and delaying healing_____

(ix) A person who has inhaled toxic gas can be immediately helped by moving him/her to open fresh air_____

(x) In providing the first aid to a victim of snake bite, lay him down with the bitten part lower than the heart._____

4. (a) What does the term apparatus mean?_____

_____

_____

(b)Mention any three(3) apparatus which can only be in biology laboratory

(i) ……

(ii) ……

(iii) ……

(c) Draw and label the diagram of hand lens

5. List five causes of nose bleedings

(i) ……

……

(ii) ……

……

(iii) ……

……

(iv) ……

……

(v) ……
……

(b) Which hazard symbol would you sue for.

(i) An acid _____
(ii) An alcohol_____
(iii) A bacterial culture_____

6. (a) What units would you use to measure your
I) body temperature _____
ii) Pulse rate _____
(b) Define shock_____
_____
(c) list any four sign of shock
(i) ……
……
(ii) ……
……
(iii) ……
……
(iv) ……
……

(d) State one function of each of following items found in the First Aid Kit

(i) Splints_____
_____
(ii) Gentian violet solution (GV)
_____
_____
(iii) Sterile gauze ----------
_____
_____

7. State the type of accident that will be prevented in each of the following practices?
a) Clear the bushes around the house_____
b) Store paraffin and detergents in their right containers_____
c) Use a sturdy ladder in climbing to the roof_____
d) Don't eat in the laboratory_____
e) Don't play in the laboratory_____
f) Don't smoke in the bed_____
g) Don't blow the nose too hard so as to remove the dried mucus_____
h) Ensure that water stored in buckets and other containers are covered with tight lids._____

i) Don't hang the clothes in the fire places_____

j) Never use the laboratory vessel to drink water _____(10 marks)

SECTION C (10 MARKS)

One student argued that the car performs the following processes which can be compared to the characteristics of living things (shown in brackets)

A- Ejects waste gases (excretion)

B- Consumes fuels (nutrition)

C- Moves (movement)

D- Rings alarms on its ours (sensitivity)

E- Its tires increase in size when inflated (Growth)

Briefly describe how the processes performed by the car differ from those characteristics of living things compared to them.

All the best